

IEEE VR 2023 Workshops

Workshop: IEEE VR KELVAR Workshop: K-12+ Embodied Learning through Virtual and Augmented Reality (8th Annual Workshop)

ORGANIZERS:

Steven Cutchin
Boise State University, USA

Erica Southgate
University of Newcastle, Australia

Jerry Alan Fails
Boise State University, USA

Manoela Milena Oliveira da Silva
Federal University of Pernambuco, Brazil

ABSTRACT:

K-12+ (K-12 and higher ed) education is currently undergoing a technological revolution creating opportunities for Virtual-, Augmented-, and Mixed-Reality based learning (hereafter referred to as XR [extended reality]) technologies. Technology integration will continue to increase as mobile devices penetrate all socioeconomic strata, and as XR technologies become affordable to schools, vocational education providers, universities, and informal educational settings. These technologies have the potential to facilitate effective learning by: developing the ability to engage students of all ages with interactive 3D simulations of real-life and artificial phenomena; presenting information that is spatially – and temporally – integrated with real objects; leveraging whole-body motions to depict and reinforce learning content. However, there are many questions about the integration of such experiences into the

classroom, such as: What curriculum topics might be addressed through XR technologies?; What socio-cultural, psychological and physiological mechanisms underlie embodied cognition?; How can we design experiences that are appropriate for the different stages of human development?; How will pedagogical approaches be influenced by such technologies?; and, What are the ethical and legal implications of using XR technology for education?

In this workshop we aim to bring together educators, developers and researchers who are interested in creating and deploying XR technologies for the educational contexts of the future. The workshop will enable participants to discuss and engage with different approaches for designing and integrating XR technologies with a specific focus on the challenges and potential for embodied learning in the classroom for K-12, vocational and higher education.

ABOUT THE WORKSHOP:

We welcomed preliminary research results and thought-provoking position and demo papers, on topics related to XR learning for K-12+:

- VR, AR & MR Technologies and Applications for formal and informal education sites (e.g.: classrooms, libraries, museums, afterschool programs)
- Theories and Applications of Embodied Cognition and Learning
- Curriculum-based Educational Applications
- Student-Teacher Relationships and Pedagogical Implications
- Classroom Integration of Technology
- Teacher and Student Content Authoring Tools

Accepted submissions cover a broad range of topics with accepted submissions ranging from theoretical position papers to hands-on learning case studies particularly in STEM fields and safety. Included accepted contributions include (with the titles and authors):

- *Is Immersive Virtual Reality in K-12 Education Ready for Primetime? Challenges, Possibilities, and Considerations* by Christos Gkoumas and Lisa Izzouzi
- *An Exploratory Study on Learning in 3D Multi-user Virtual Environments* by Ling Zhang, Yingmei Guo, Qiaodi Ma, and Yun Zhou

- *Importance of Democratization for Virtual Reality in Education* by Katrien Jacobs
- *Engaging Novice VR Users to VR for STEM Education: Enhancing User Comfort, Health and Safety in VR and AR* by Aleshia Hayes, Danita Bradshaw-Ward, Erin Howard, Megan McAdams, and Debbie Cockerham
- *Interactive Cues on Geometry Learning in a Virtual Reality Environment for K-12 Education* by Qian Chen, Ling Zhang, Bingyu Dong, and Yun Zhou
- *An Immersive Geometry Environment for Mathematics Education: Taxonomy and Preliminary Evaluation* by Lui Albæk Thomsen, Niels Christian Nilsson, and Rolf Nordahl
- *Ohm VR: solving electronics escape room challenges on the roadmap towards gamified STEAM education* by Piotr Zamojski, Norbert Barczyk, Marek Frankowski, Artur Cybulski, Konrad Nakonieczny, Marek Makowiec, Magdalena Igras-Cybulska
- *Children's evaluation of a virtual reality boating safety education resource* by Erica Southgate.
- *Case study: Strategy Business Administration Interactive XR Learning* by Jonathan Giron, Ayala Berenson, and Niron Hashai

OBJECTIVES:

While XR technologies have existed for years, they will only continue to expand in terms of how they impact everyday users. With

their continued growth, it is imperative to constantly investigate their learning potential and impact.